MUHAMMAD ABDULLAH KHAN

Full-Stack Developer · AI Application Developer

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PROFESSIONAL SUMMARY

Full-stack developer with 1.5 years of professional experience shipping production systems across the **MERN stack, Next.js, NestJS and TypeScript**, now focused on **AI application development** — building RAG pipelines, LLM tool-calling workflows and multi-channel conversational agents on OpenAI and OpenRouter. Delivered a Twilio-based WhatsApp automation platform where clients connect their own number to an AI chatbot. Comfortable owning features end to end: schema design and query tuning, REST/WebSocket APIs, front-end, and CI/CD on Fly.io, Vercel and Netlify.

TECHNICAL SKILLS

AI / GenAI: OpenAI API, OpenRouter, RAG (retrieval-augmented generation), Vector Embeddings, AI Agents & Tool/Function Calling, Voice AI Agents (Retell), Workflow Automation (n8n), Prompt Engineering

Backend: Node.js, NestJS, Express.js, TypeScript, REST APIs, WebSocket / Socket.io, JWT & RBAC, Twilio (WhatsApp / SMS)

Frontend: React.js, Next.js, JavaScript, TypeScript, Tailwind CSS, Bootstrap, HTML5, CSS3, Responsive Design

Databases: PostgreSQL, MySQL, TypeORM, Sequelize, Indexing, Triggers, Query Optimization

DevOps: Git, GitHub Actions, CI/CD Pipelines, Docker, Fly.io, Vercel, Netlify, Render, Railway

Languages: JavaScript, TypeScript, Python

PROFESSIONAL EXPERIENCE

Associate Software Engineer | **Iris Labs** *Mar 2026 – Present*

- Built a **multi-tenant WhatsApp automation platform** where clients connect a purchased number to their own AI chatbot via **Twilio**, with per-client prompts, memory and knowledge bases on **OpenAI / OpenRouter**.
- Implemented **RAG pipelines** and **LLM tool-calling workflows** so chatbots answer from client-specific data and trigger backend actions (lookups, bookings, record updates) from natural language.
- Built **voice AI agents with Retell** and orchestrated multi-step automations in **n8n**, connecting the AI layer to external services.
- Architected backend services in **NestJS + TypeScript** with TypeORM/PostgreSQL, JWT & RBAC, and type-safe DTO validation.

MERN Stack Developer | **Waywe Gaming — Lahore, Pakistan** *Apr 2025 – Mar 2026*

- Developed a full-stack **HRM system** from scratch with role-based access using React, Node.js and Sequelize.
- Built a **real-time game backend** with Socket.io and REST APIs, plus a React dashboard for role-based game statistics.
- Migrated backend services to **Fly.io** and implemented **CI/CD pipelines with GitHub Actions**, cutting manual deployment effort.
- Integrated **deep linking** for the referral system and Facebook authentication.

MERN Stack Intern | **A4H — Bolton, UK (Remote)** *Sep 2023 – Jan 2024*

- Built responsive React.js interfaces and backend APIs with Node.js and Express.js for data handling and secure authentication.
- Worked in an agile team, delivering on deadline across an international remote setup.

KEY PROJECTS

Receta AI — AI Appointment & Triage Platform for Hospitals — *NestJS, React, Supabase, OpenAI, Twilio, Stripe, Zoom, RAG*

- Built a platform that **onboards hospitals** to create their own AI bot: admins pick the model, set instructions, attach a **RAG knowledge base**, and connect a **Twilio** number so patients get hospital info over WhatsApp or SMS.
- Designed **AI triage workflows** on **OpenAI** models that interpret a patient's query, route them to the right specialist based on doctor availability, and book the appointment automatically.
- Integrated **Stripe** for appointment payments, **Zoom** for auto-generated online consultations, and notifications to the doctor and hospital on each booking.
- Developed the full stack in **NestJS + React with Supabase**, enabling patient–doctor connection for follow-up queries.

Reab — Ethical, Privacy-First Social Media Platform *(reab.com)* — *React, Node.js, Socket.io*

- Contributed to a social media platform with **real-time messaging** (Socket.io) and **video content management** for posts, feeds and communities.
- Built a **referral system with earnings**, rewarding users for successful referrals across the platform.

E1 — Creator & Brand Affiliate Marketing Platform *(e1.app)* — *Next.js, NestJS, PostgreSQL, Stripe*

- Built a two-sided **affiliate marketing platform** onboarding creators and brands: brands register campaigns, assign creators, and creators promote each campaign on their own social media.
- Aggregated and displayed **post-level insights and reach** for each creator against a campaign, giving brands a clear view of performance and audience reach.
- Supported **subscription-based and day-based campaign models** with flexible plan handling.
- Integrated **Stripe** for payments: a fixed payout is released to the creator on campaign completion and withdrawn directly via Stripe.

PokerFace — Real-Time Multiplayer Poker Game Backend — *Node.js, Socket.io, PostgreSQL, Sequelize, WebSockets*

- Built the complete **real-time backend server** for a multiplayer poker game using **WebSockets (Socket.io)** for live game state and player actions.
- Designed the full game logic and **REST APIs**, backed by a **PostgreSQL** relational database with **Sequelize ORM**, supporting multiple concurrent users per table.

EDUCATION & TRAINING

BS in Computer Science | **University of Central Punjab, Lahore** *2021 – 2025*

Software Trainee | **Integriti — Lahore, Pakistan** *Feb 2025 – Apr 2025* · Built React.js interfaces and Node.js/Express APIs handling auth and business logic.